This quiz has six problems worth 10 points.

Below is the matrix A and its reduced row echelon form, R. Use these to answer the following questions.

$$A = \begin{bmatrix} 2 & -2 & 4 & 3 \\ 4 & -8 & 10 & 7 \\ 6 & 14 & 2 & 4 \end{bmatrix} \qquad R = \begin{bmatrix} 1 & 0 & 3/2 & 5/4 \\ 0 & 1 & -1/2 & -1/4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

1. (1 pt) What is the rank of A?

2. (1 pt) What are the *free variables* in this case? (Assume $\mathbf{x} = (x_1, x_2, x_3, x_4)$.)

3. (2 pts) For each free variable, find a *special solution*.

4. (2 pts) Efficiently and precisely describe $N(A)$, the null space of A.

$$A = \begin{bmatrix} 2 & -2 & 4 & 3 \\ 4 & -8 & 10 & 7 \\ 6 & 14 & 2 & 4 \end{bmatrix} \qquad R = \begin{bmatrix} 1 & 0 & 3/2 & 5/4 \\ 0 & 1 & -1/2 & -1/4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

5. (2 pts) Efficiently and precisely describe $C(A)$, the column space of A. (Note that A and R are copied above.)

6. (2 pts) Below is the augmented matrix $[A\ \mathbf{b}]$ for the system of equations $A\mathbf{x} = \mathbf{b}$ and its reduced row echelon form, R. Find the complete solution to the system $A\mathbf{x} = \mathbf{b}$.

$$A = \left[\begin{array}{cccc:c} 2 & -2 & 4 & 3 & 6 \\ 4 & -8 & 10 & 7 & 8 \\ 6 & 14 & 2 & 4 & 38 \end{array}\right] \qquad R = \left[\begin{array}{cccc:c} 1 & 0 & 3/2 & 5/4 & 4 \\ 0 & 1 & -1/2 & -1/4 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$